

K-Nearest Neighbors (KNN)

Eager and lazy learning

Learning system can be characterized in a number of ways. One way is based on when the bulk of the effort is applied.

- **Eager Learning** -- In eager learning, more effort is applied to training (i.e., learning) than is applied to learning new data.
learning.
- **Lazy Learning** -- In lazy learning, learning is delayed until new data is to be classified (or a regression parameter is to be estimated).

Parametric and non-parametric learning

Another way is based on whether or not parameters are estimated during the training phase for later use in characterizing new data.

- **Parametric Learning** -- In parametric learning, parameters are estimated during the training phase for later use in characterizing new data.

- **Non - parametric Learning** -- In non-parametric learning, parameters are not estimated during the training phase for later use in characterizing new data.

K-nearest neighbors algorithm

The K-nearest neighbors algorithm doesn't learn a discrimination function for the training data for later use, but rather just memorizes the training dataset. It is a lazy, non-parametric algorithm and is additionally referred to as an example of **instance - based learning**.

K-nearest neighbors algorithm

The K-nearest neighbors algorithm is as follows:

A) Set up

- 1) Store the training data.
- 2) Choose a value for k .
- 3) Choose a distance metric.

B) New sample data

- 1) Find the k -nearest neighbors of the sample data $\mathbf{x}_i$ in the training set $\{\mathbf{x}_{training}\}$.
- 2) Assign the category of $\mathbf{x}_i$ using weighted majority voting among the k -nearest neighbors. Two typical weightings are uniform and inversely proportional to the distance between the sample data and each of its k -nearest neighbors.

K-nearest neighbors algorithm issues

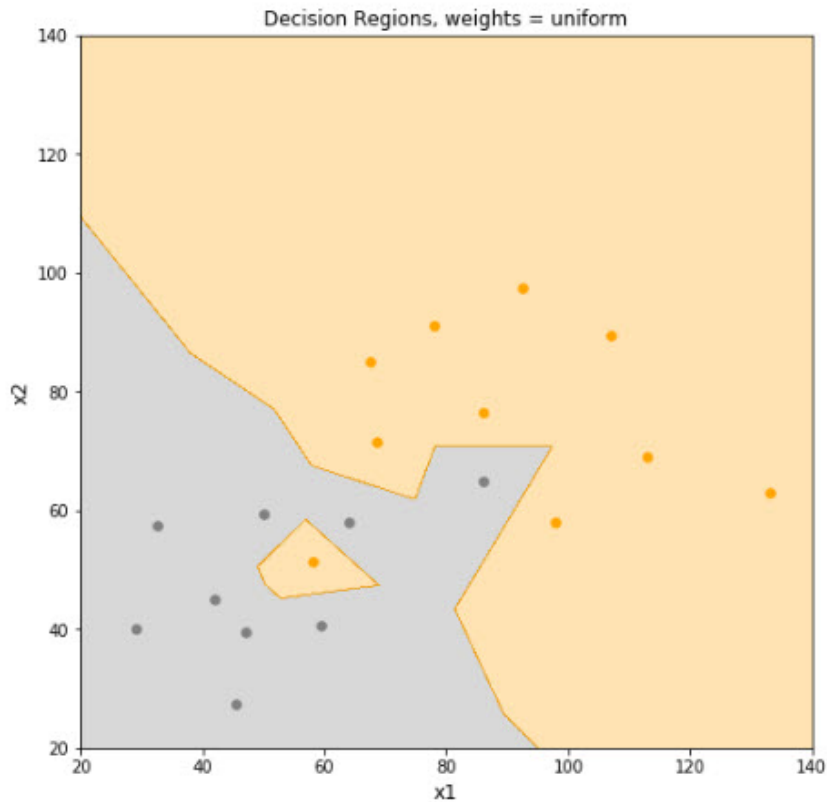
The K-nearest neighbors algorithm has the following issues:

- Subject to both underfitting and overfitting.
- Computational complexity grows linearly in the the number of data points and the number of features.
- The choice of the distance metric is important and may not be obvious.

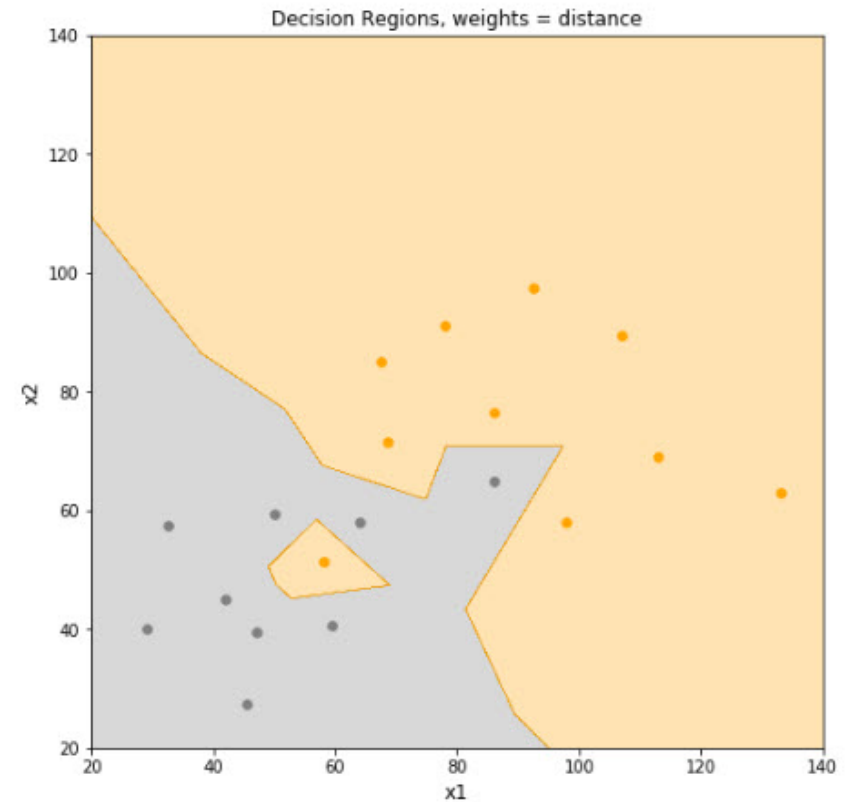
Reshow the KNN algorithm (slide 5)

K-neighbors = 1

Neighbors = 1 , weights = uniform

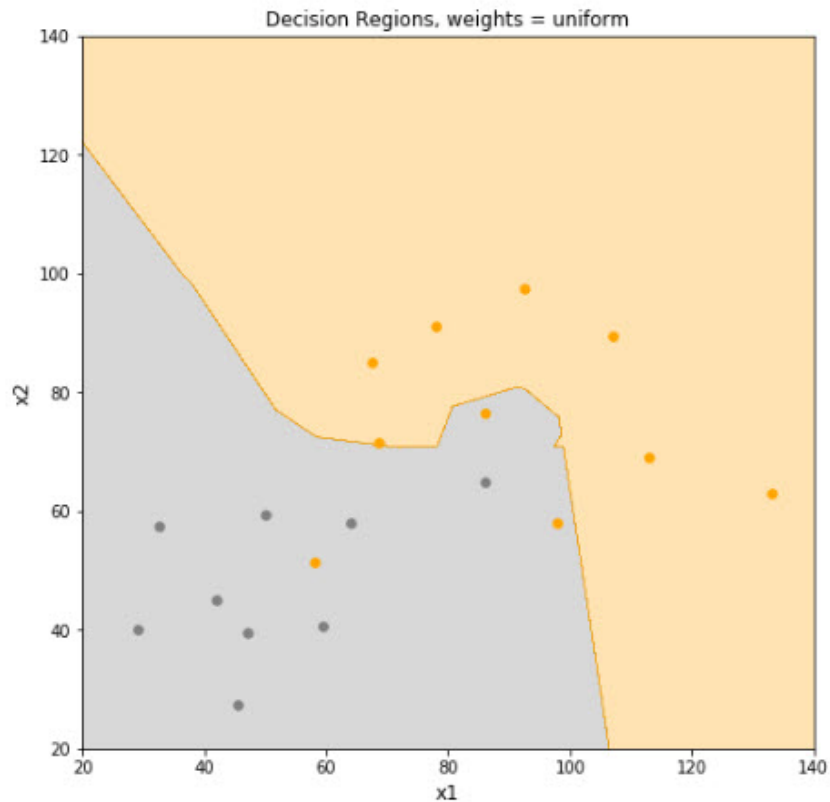


Neighbors = 1 , Weights = distance

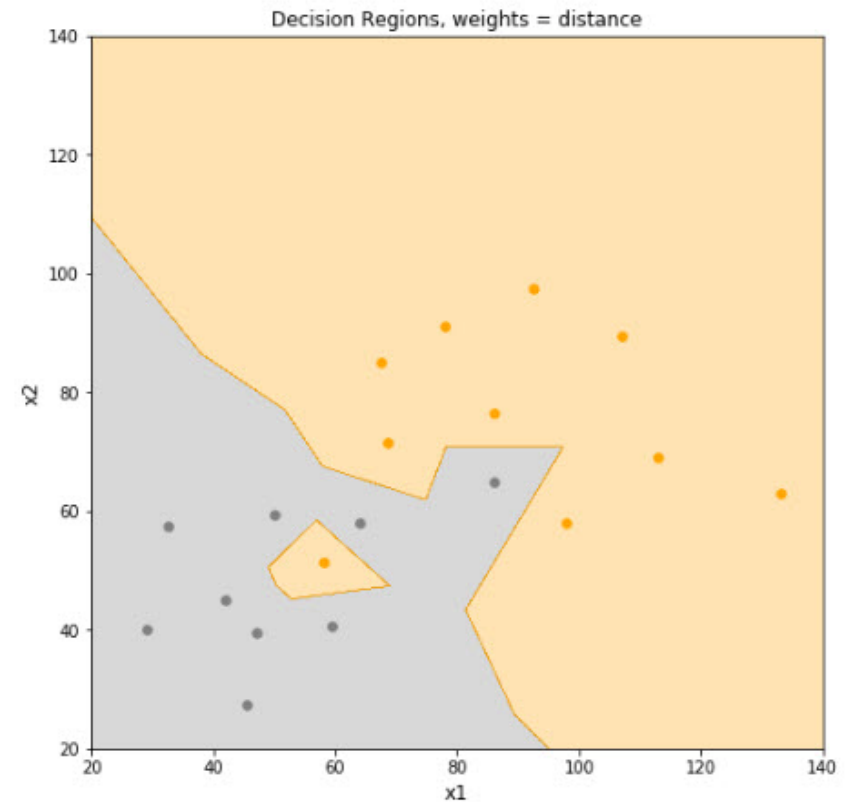


K-neighbors = 2

Neighbors = 2 , weights = uniform

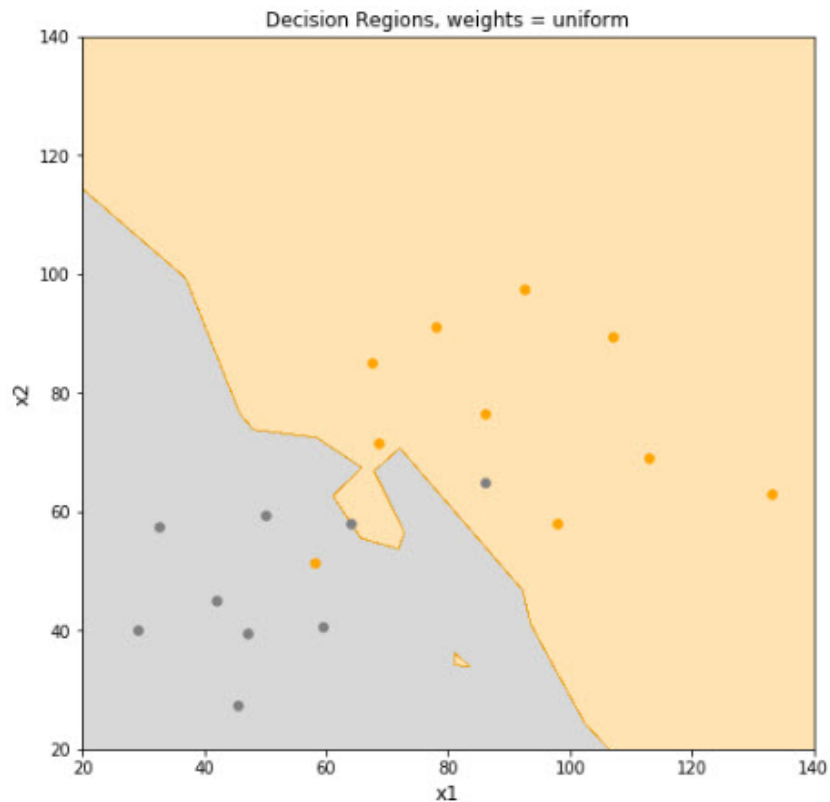


Neighbors = 2 , Weights = distance

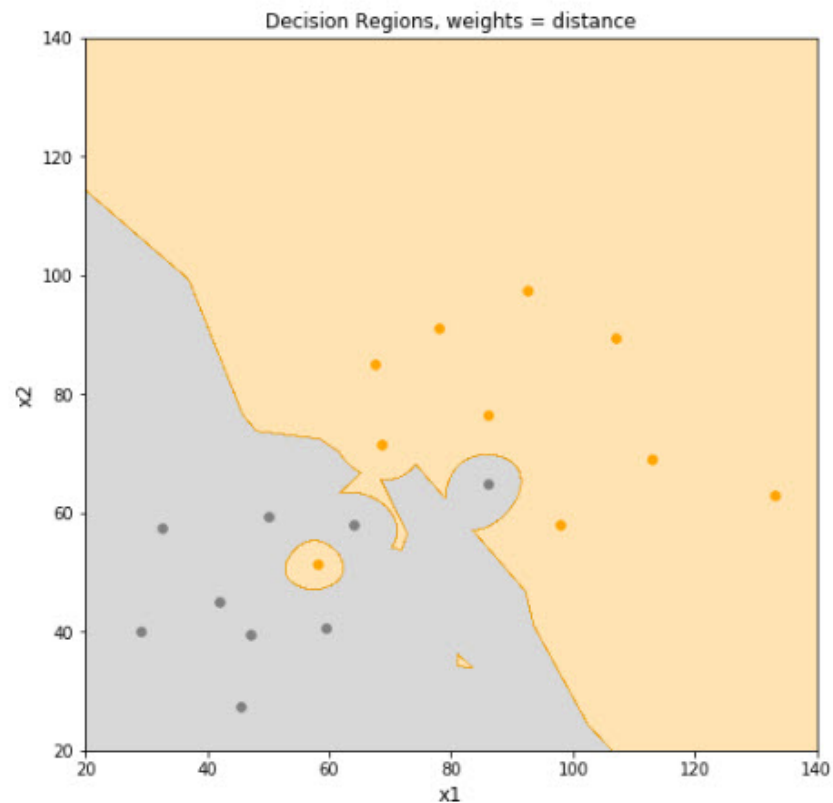


K-neighbors = 3

Neighbors = 3 , weights = uniform

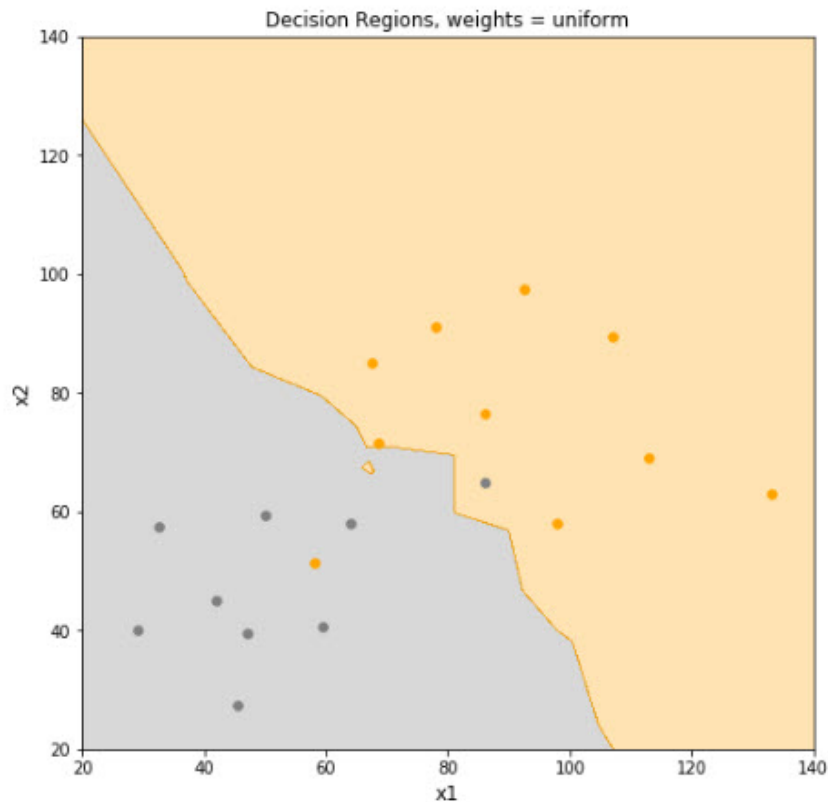


Neighbors = 3 , Weights = distance

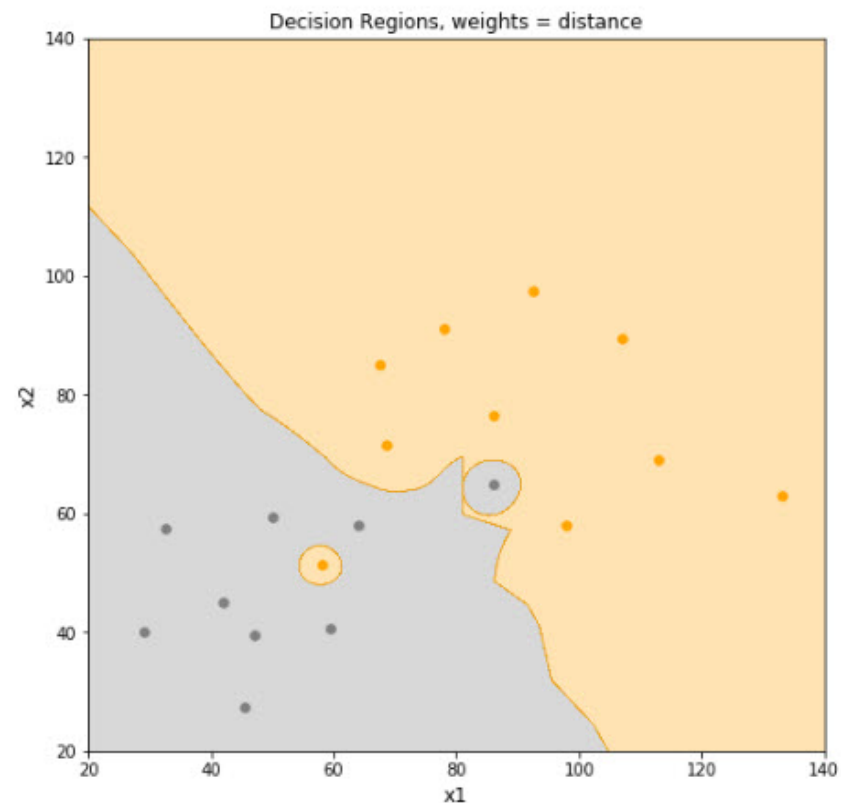


K-neighbors = 4

Neighbors = 4 , weights = uniform

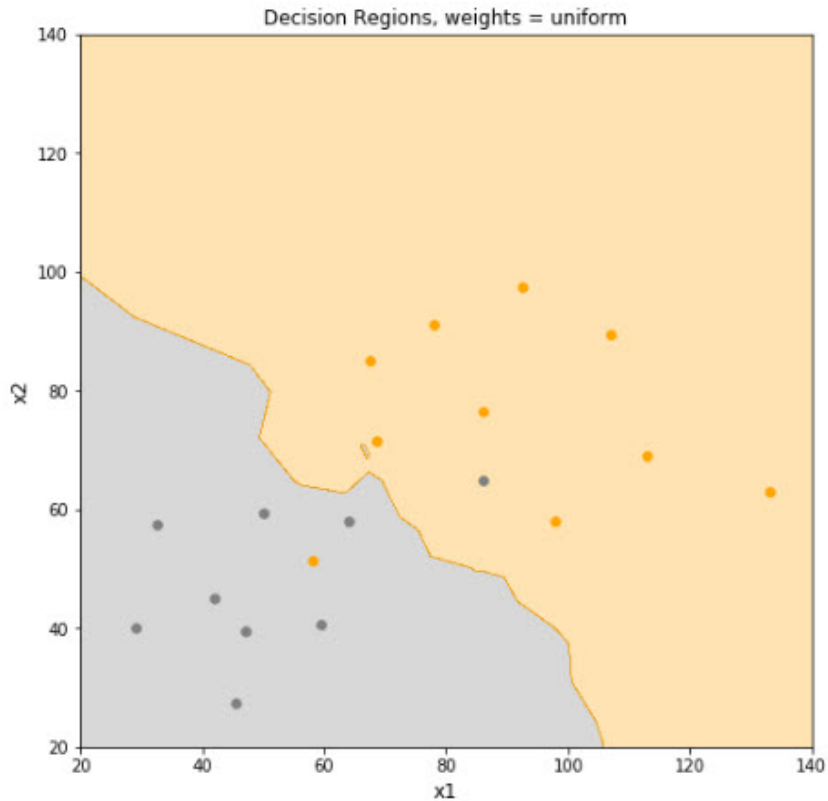


Neighbors = 4 , Weights = distance

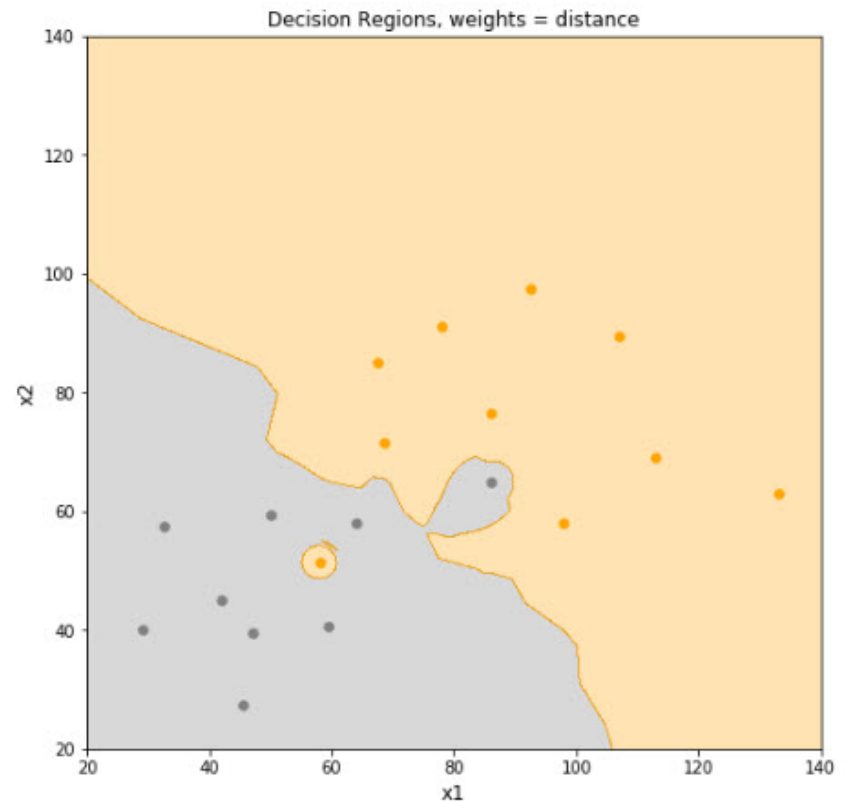


K-neighbors = 5

Neighbors = 5 , weights = uniform

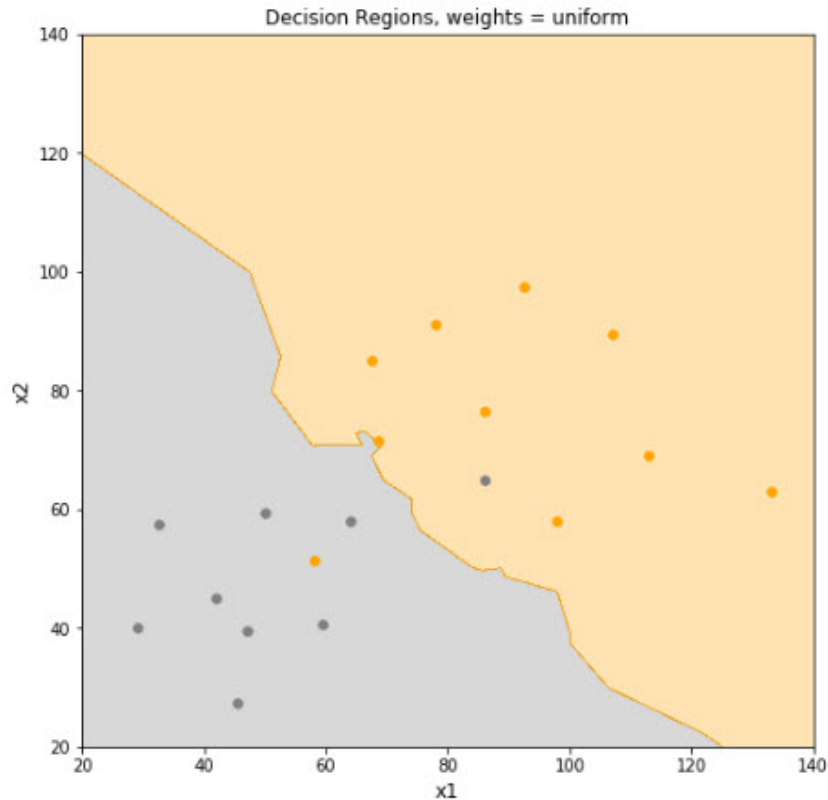


Neighbors = 5 , Weights = distance

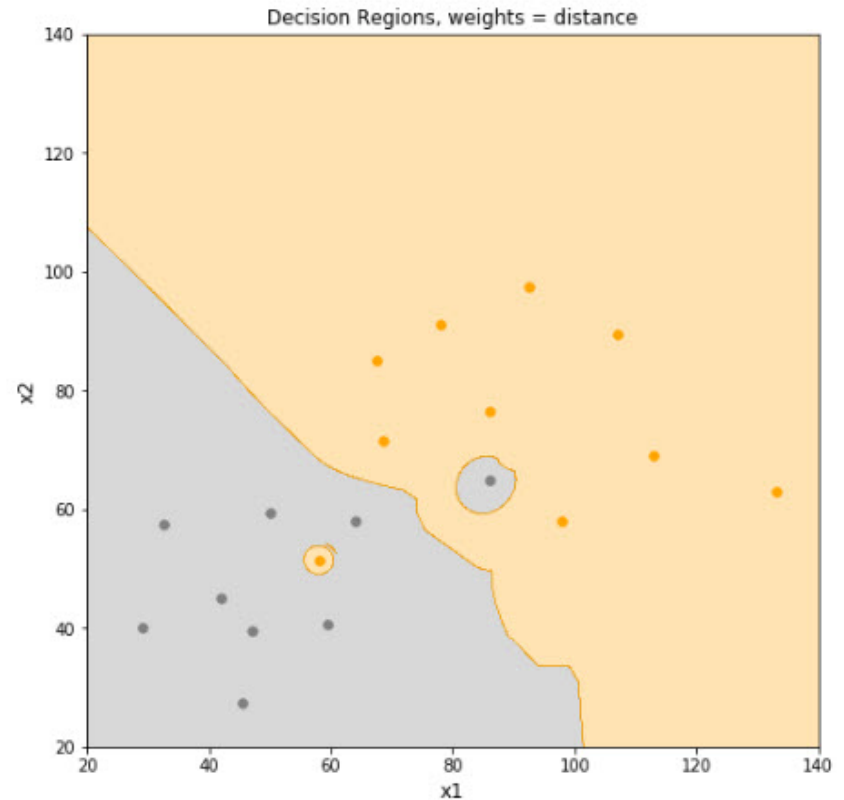


K-neighbors = 6

Neighbors = 6 , weights = uniform

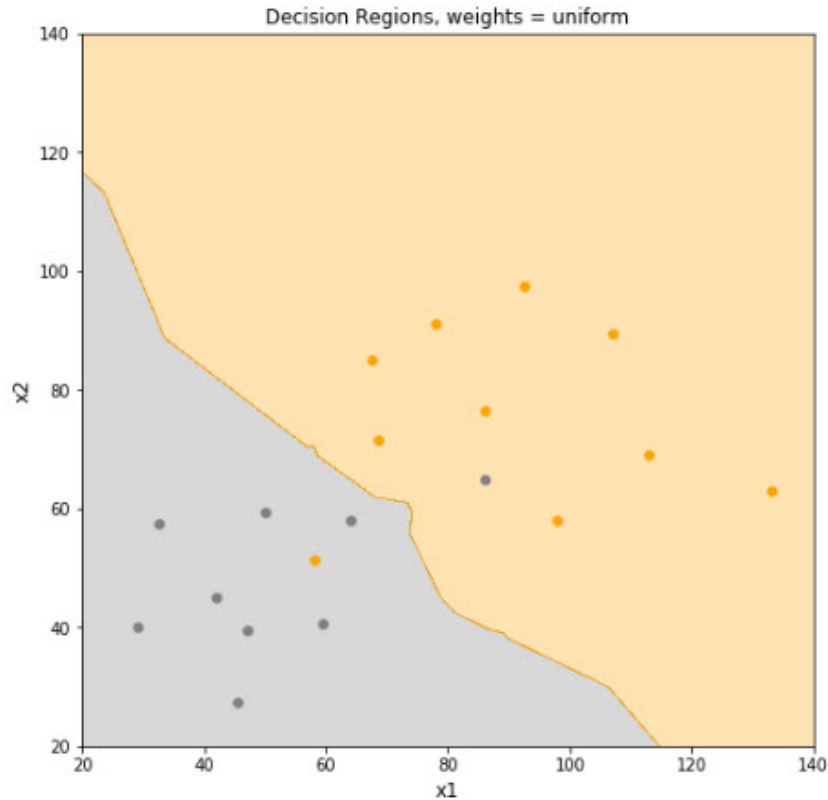


Neighbors = 6 , Weights = distance

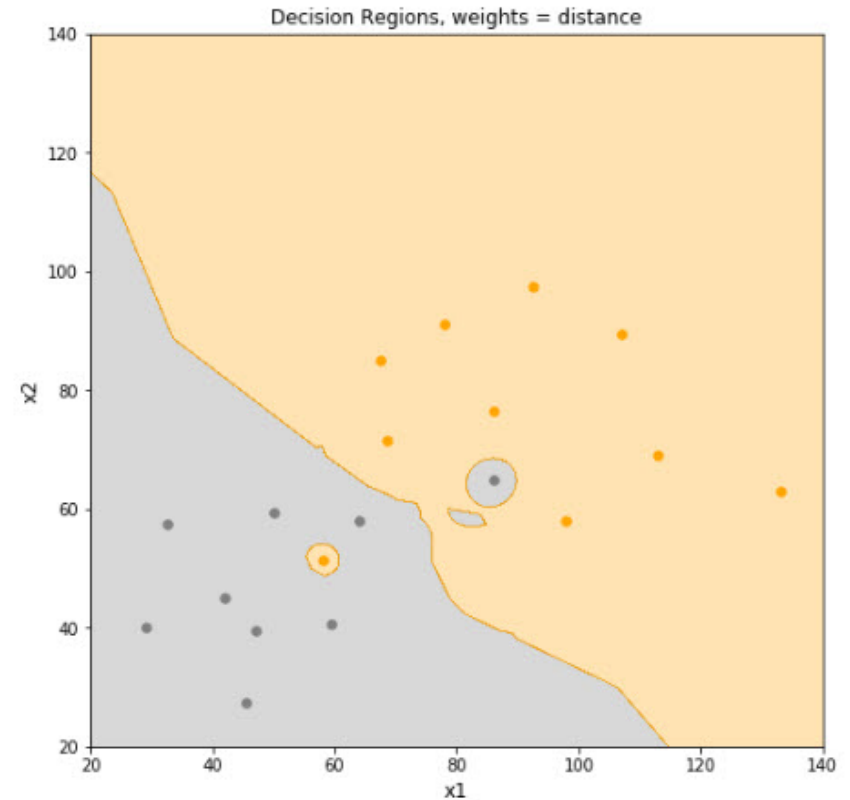


K-neighbors = 7

Neighbors = 7 , weights = uniform

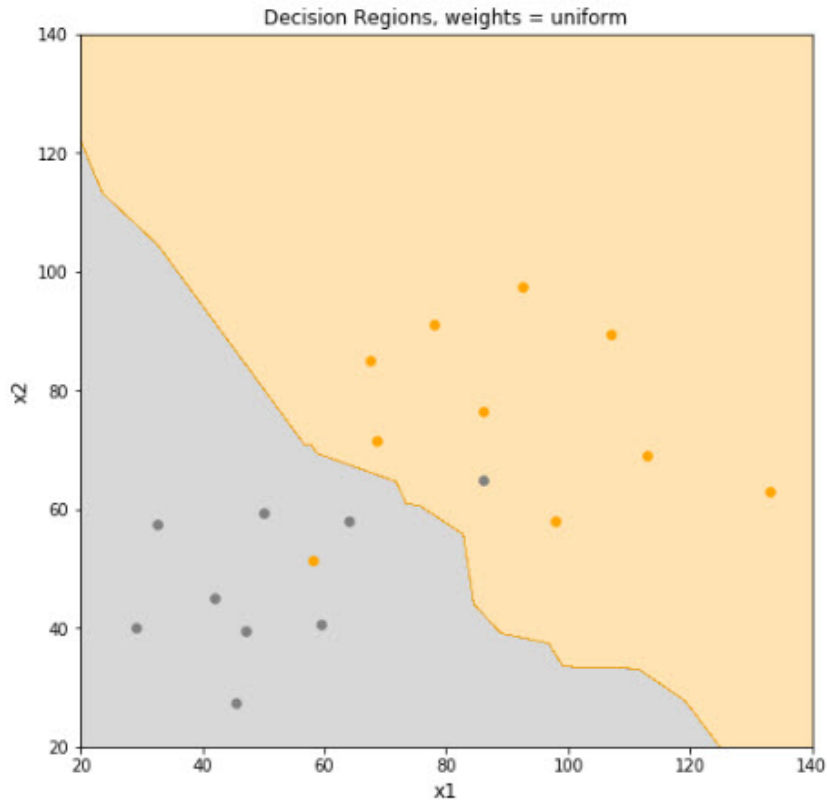


Neighbors = 7 , Weights = distance

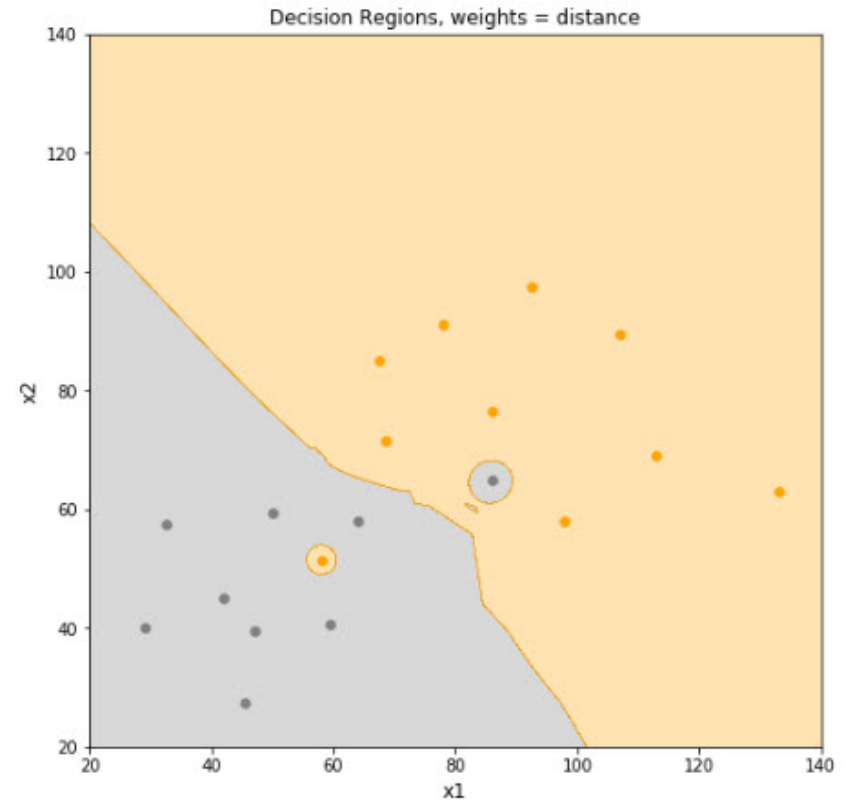


K-neighbors = 8

Neighbors = 8 , weights = uniform



Neighbors = 8 , Weights = distance



Scikit-learn K-Neighbors Classifier class

class sklearn.neighbors.KNearestNeighborsClassifier (...)

Parameter	Default
n_neighbors	5
weights	'uniform'
algorithm	'auto'
leaf_size	30
p (power parameter for metric)	2
metric	'minkowski'
metric_params	None
n_jobs	None

Scikit-learn K-Neighbors Classifier methods

Method	Description
<code>fit(X, y)</code>	Fit the model using the training set (X, y).
<code>get_params([deep])</code>	Get parameters for this estimator.
<code>kneighbors([X,n_neighbors, return distance])</code>	Finds the K-neighbors at a point.
<code>kneighbors_graph([X,n_neighbors, mode])</code>	Computes the (weighted) graph of k-neighbors for points in X.
<code>predict(X)</code>	Predict the class labels.
<code>predict_proba(X)</code>	Returns probability estimates for the test data X.
<code>score(X, y[, sample_weight])</code>	Returns the mean accuracy on the given test data and labels.
<code>set_params(**params)</code>	Set the parameters of this estimator.